

BrickAtlas Hierarchy: Crossed Evaluation of Brick Structure, Reasoning, and Executable Intervention

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Abstract

Brick reasoning involves both object complexity and task composition. Increasing the number of views or edits does not create independent objects, and naming a task “physical” does not validate its mechanics. We introduce Hierarchy-2, a public development release of BrickAtlas crossing 40 original layouts with 35 shared task families. Four structural strata contain ten layouts each; every layout supports 18 atomic, ten metacognitive/physical, four procedural, and three integrative tasks, for 1,400 questions on 5,070 visible parts. A state-transition interpreter checks submitted actions, while resource and possible-world validators evaluate scheduling and conditional policies. Rapier supplies actual collision queries and sampled dynamic traces for all objects. The original BrickAtlas website now exposes every layout as interactive 3D geometry, with exploded inspection and playback of submitted actions. A fixed 16-question low-cost pilot yields 14 exact successes for GPT-4.1 mini at \$0.011304. This establishes pipeline functionality, not calibrated difficulty or a model ranking. Quality auditing finds only 17/40 layouts within the declared nominal-drift tolerance, which limits the physical claims of the release.

1. Introduction

Digital brick environments offer discrete object identities, hierarchical assemblies, and explicit edits. These properties support reproducible evaluation across perception, structural reasoning, and intervention. However, benchmark growth can obscure rather than resolve a model’s failure: more derivatives of a few objects create larger denominators without broader source coverage. Likewise, a final correct object does not show that an agent constructed, repaired, or inspected it correctly.

BrickAtlas organizes evaluation along two axes. The vertical axis describes the structural scope of an object: component, assembly, mechanism, or system. The horizontal axis describes task composition: atomic operations, metacognitive and physical reasoning, executable procedures, and in-

tegrated constraints. Crucially, the same 35 families occur at every level, so comparisons can hold task identity fixed. Figure 2 shows the complete crossed design.

This release extends the 18-layout Hierarchy-1 library with 22 original compositions, preserves its six Mechanism-1 objects, and replaces incomplete cross-level task coverage with 1,400 questions. These are 40 layouts, not 40 independently sampled mechanical families: new layouts reuse authored components. OMR/LDraw examples inform visual design but are not benchmark cases. The retired 5,120 procedural sources remain historical regression data and are excluded from the new scale.

Three implementation changes improve inspectability. First, all physics-derived questions use archived Rapier observations rather than handwritten response tables. Second, action answers execute a public state machine; an equivalent legal order can pass without matching a single reference list. Third, dataset geometry is available inside the original application, allowing inspection of modules, individual parts, exploded configurations, and intermediate action states.

We make the following contributions:

1. a balanced four-by-four organization with 40 authored layouts and 35 task families repeated on every layout;
2. executable action, schedule, and possible-world contracts with negative controls and explicit physical-evidence boundaries;
3. an integrated 3D browser sharing model data with the offline release and displaying reference or submitted programs; and
4. a fully receipted low-cost pilot and a per-object quality audit that reports unstable simulations rather than hiding them.

2. Related Work

BrickGPT [7] addresses stable brick generation; BrickNet [3] uses typed connection graphs. BrECS [1] studies sequential assembly with budgets. Hierarchy-2 evaluates an existing design and its interventions; it does not claim comparable asset scale or a new generative method.

Level	Layouts	Parts/layout	Questions
D1	10	14–28	350
D2	10	45–126	350
D3	10	43–154	350
D4	10	214–437	350

Table 1. Actual release counts. D2 and D3 overlap in part count because the strata describe structural scope, not measured psychometric difficulty.

LEGO-Puzzles [8] and PhyBlock [5] cover visual, spatial, ordering, and counterfactual reasoning. Manual-based assembly methods such as MEPNet [10], Neural Assembler [11], and TreeSBA [2] recover poses or construction steps. Our contribution is a shared task-by-source organization with executable contracts, rather than a claim that each individual family is new.

Break and Make [9] evaluates interactive disassembly and reconstruction. WorkBenchMark [6] and BC-Bench [4] address embodied or compositional assembly. We explicitly separate digital state validity from real insertion force, connector tolerance, grasping, and hardware transfer.

3. Library and Crossed Design

3.1. Four Structural Strata

D1 contains local devices such as gimbals, grippers, hatches, and valve stands. D2 introduces complete assemblies including service lifts, loaders, trackers, and inspection craft. D3 contains articulated whole mechanisms such as cranes, observatories, robots, and excavators. D4 adds multi-workcell systems, including cargo terminals, docking yards, energy stations, and service hangars.

Table 1 reports actual counts rather than nominal target intervals. The library has 355 rigid modules and 318 declared joints. All layouts are agent-authored using reusable construction primitives; no independent human design study was conducted. Canonical geometry checks reject exact duplicates after removing names and colors, but this does not establish semantic independence.

3.2. Representation and Provenance

A part has a shape, size, local position, quaternion, color, and module membership. Modules have world poses, masses, roles, and anchoring status. Joints identify end-points, local anchors, type, axis, and limits. The visible vocabulary includes bricks, plates, beams, windows, slopes, arches, cylinders, wheels, axles, gears, and spheres.

These shapes are LEGO-like digital primitives, not verified commercial part specifications. Original-site model JSON and offline model JSON are byte-identical. Each object also records its family and lineage; future train/test splits must group related layouts and reused components.

The present public development set has no hidden confirmation split.

3.3. Matched Coverage

Every level has 350 questions. Atomic tasks are repeated across levels, including recognition, counting, local edits, topology, and pose corrections. Meta tasks include both physical-record interpretation and uncertainty-aware decisions. The four procedural families are assembly, disassembly, supported service repair, and compound editing. The integrative families are distributed repair, resource scheduling, and inspection-conditioned policies.

Task identifiers bind source and family. Changing a render, color, answer order, or fault creates a condition within a source, not a new source object. Although all families are crossed, many local questions remain easy with explicit structured input; a larger scene alone does not guarantee a harder local operation.

4. Task and Scoring Contracts

4.1. Atomic and Metacognitive Questions

Atomic tasks return a choice ID. Each distractor changes a relevant field, set, count, identity, or transformation. Option rotation is deterministic per task, and duplicate option values are removed. Some questions have fewer than four distinct alternatives; chance baselines must use the actual option count.

Multi-answer tasks require the entire feasible set. In accessibility, an empty feasible set is valid if every proposed path collides. Counterfactual support excludes the explicitly removed object and recomputes reachability to any anchored root through remaining joint edges, allowing redundant support paths.

Inspection uses a finite uniform prior and deterministic observations. For query q with observation partition sizes n_o over N worlds,

$$U(q) = \frac{-\sum_o (n_o/N) \log_2(n_o/N)}{c(q)}. \quad (1)$$

All utility-maximizing queries are accepted. Abstention tests whether all compatible worlds imply the same action. Posterior tasks condition the finite prior on an observed query result. Pareto tasks return every non-dominated alternative under cost, mass, and stiffness.

Public inputs explicitly disclose structural metadata where required. Therefore most atomic scores measure structured reasoning with auxiliary images, not isolated visual perception. Image-removal, symbolic-removal, and viewpoint ablations are necessary before claiming visual reasoning gains.



Figure 1. **Representative layouts at all four structural levels.** Rows progress from D1 to D4; three fixed representatives per level are shown. These are canvas renders from the original application, not generated concept images. All 40 layouts and all four views per layout are provided in the supplementary index.

4.2. Executable Procedures

An action is a tuple

$$a = (P_a, F_a, A_a, D_a, c_a), \quad (2)$$

with required facts, forbidden facts, add effects, delete effects, and cost. From state S , execution requires $P_a \subseteq S$ and $F_a \cap S = \emptyset$, then applies

$$S' = (S \setminus D_a) \cup A_a. \quad (3)$$

The interpreter checks every action in submitted order, stops at the first invalid transition, and records its cause. Success requires all goal facts and the declared budget.

Assembly places a module only after its declared predecessors. Disassembly removes children before parents. Repair sequences support, open, remove, replace, verify, close, and release an interface. Compound editing substitutes a

color operation within a protected service sequence. Independent actions can commute. Duplicate placements, unknown IDs, missing terminal steps, or budget overruns fail.

The transition contract certifies symbolic intervention rules. It does not automatically simulate robot motion or tool contact for every repair action. The website changes module visibility and colors according to these same transitions and displays state facts alongside the 3D replay, so presentation and scoring share one interpreter.

4.3. Integrated Constraints

Distributed repair combines multiple independent faulty modules under one budget and one final goal. Scheduling requires integer start times, predecessor completion, non-overlapping use of exclusive workcells, and completion by a deadline. Inspection policies select a query and map every possible observation to an action; the validator checks

Hierarchy-2: structural scope × task composition

	Atomic	Meta / physics	Execution	Integration
D1 Components	180 questions 10 sources / level	100 questions 10 sources / level	40 questions 10 sources / level	30 questions 10 sources / level
D2 Assemblies	180 questions 10 sources / level	100 questions 10 sources / level	40 questions 10 sources / level	30 questions 10 sources / level
D3 Mechanisms	180 questions 10 sources / level	100 questions 10 sources / level	40 questions 10 sources / level	30 questions 10 sources / level
D4 Systems	180 questions 10 sources / level	100 questions 10 sources / level	40 questions 10 sources / level	30 questions 10 sources / level

Figure 2. **Crossed structural scope and task composition.** Each cell contains ten sources. Every source contributes 18 atomic, ten meta/physical, four procedural, and three integrative questions. The axes are design strata; measured monotonic difficulty is not assumed.

all worlds instead of only a sampled rollout.

No LLM judge determines correctness. Empty or malformed outputs remain failures. A batch evaluator must retain missing cases in its frozen denominator and reject unknown or duplicate case IDs. Per-family and per-source reporting is preferred to a pooled percentage.

5. Physics and Quality Audit

Rapier 0.20.0 runs at 120Hz with 12 solver iterations. Visible parts are conservatively approximated by cuboids and grouped into rigid bodies. Fixed, revolute, prismatic, and spring descriptors instantiate the declared mechanism. Bases are anchored; holding motors use public parameters. A separate static world performs three continuous service shape casts per object, excluding the serviced module.

After 120 nominal steps, the audit records maximum body-origin drift. Only 17/40 objects have drift below 0.35 scene units. The remaining 23 are explicitly flagged in both the release and application. Drift may reflect compliant articulation, conservative collision envelopes, or defective authored geometry; this test does not establish the cause. We do not certify the complete library as physically stable.

An off-center impulse is followed by 120 simulation steps, sampled at eleven time points. Dynamic questions ask for the maximum of the *sampled translational response*, not the continuous maximum, a rotational safety bound, or

Level	Atomic	Meta	Procedure	Integrated
D1	1/1	1/1	1/1	1/1
D2	1/1	0/1	1/1	1/1
D3	1/1	1/1	1/1	1/1
D4	1/1	0/1	1/1	1/1

Table 2. GPT-4.1 mini pipeline pilot. Each cell is one fixed question. These are observed counts, not estimates of whole-cell accuracy.

a calibrated load threshold. They remain interpretable on unstable objects as diagnostic traces but cannot be used to claim real-world robustness.

Other checks cover unique identities, finite poses, positive dimensions, declared dependency order, rendered-image hashes, and exact website/ dataset parity. Inter-module bounding-box overlaps are published as diagnostics rather than silently treated as precise mesh collisions.

6. Pilot and Inspectable Release

The fixed pilot selects boundary identification, information gain, service repair, and scheduling at each level. GPT-4.1 mini receives one image and public structured evidence at temperature zero, with a 2,200-token output cap, no automatic retries, and no provider fallback. The 16 responses

GPT-4.1 mini · 16-case pipeline pilot

	Atomic	Meta / physics	Execution	Integration
D1	1/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)
D2	1/1 n=1 (pilot)	0/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)
D3	1/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)
D4	1/1 n=1 (pilot)	0/1 n=1 (pilot)	1/1 n=1 (pilot)	1/1 n=1 (pilot)

Descriptive cells only. Not population accuracy or calibrated difficulty.

Figure 3. Pilot cell counts on the two design axes. A green cell means one correct response, not established 100% population accuracy.

yield 14 exact successes at \$0.011304; raw replies, provider IDs, costs, and public-input hashes are archived. Table 2 preserves each denominator.

Deterministic checks accept all 1,400 reference answers. A separate public-precondition solver succeeds on 200 action tasks. The simple constant-choice/empty-answer control succeeds on 268/1,400 cases, with heterogeneous choice cardinalities and output contracts. Reference acceptance is evaluator validation, not learned-model performance; the action solver has privileged symbolic rules compared with a purely visual agent.

The original application’s homepage lists all 40 benchmark models. Each opens an orbit-controlled 3D workspace with zoom, pan, canonical views, module isolation, explosion, and image export. Its task panel shows all 35 contracts, and action JSON can be replayed with a step slider and first-failure feedback. The same renderer generates 160 audited model views. Every source and every task family is indexed in the supplement; full question/choice/input/answer records are shipped in the Chinese question bank and machine-readable release.

7. Limitations and Next Validation

The release substantially increases coverage and inspectability, but the structural levels remain authored categories. There is no human timing study, item-response calibration, or causal proof that higher levels are harder. Shared constructors create correlated layouts. The 16-question pilot is too small for population accuracy, robust ranking, or level-wise discrimination claims.

Physical realism is limited by compound bodies, cuboid envelopes, anchored bases, and uncalibrated material parameters. Digital primitive connectivity is not commercial LEGO clutch-force validity. Only a subset passes the

nominal-drift check. Symbolic repair transitions and exploded display layouts are not collision-free robot trajectories.

Future confirmation requires source-family-disjoint hidden designs, manual geometry review, calibrated connectors and contact mechanics, human difficulty measurements, and information-condition ablations. These are outstanding scientific requirements, not outcomes inferred from passing software tests.

8. Conclusion

Hierarchy-2 provides a common source-by-task organization for brick reasoning, execution, and integrated decisions. Its 40 layouts, 1,400 questions, state-based scoring, actual simulator records, and original-site 3D access make the benchmark easier to inspect and extend. Explicit quality failures and small-sample result boundaries remain part of the release.

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